EXP NO: 11	DESIGN AND IMPLEMENTATION OF SYNCHRONOUS AND RIPPLE COUNTER	DATE: 15/11/22

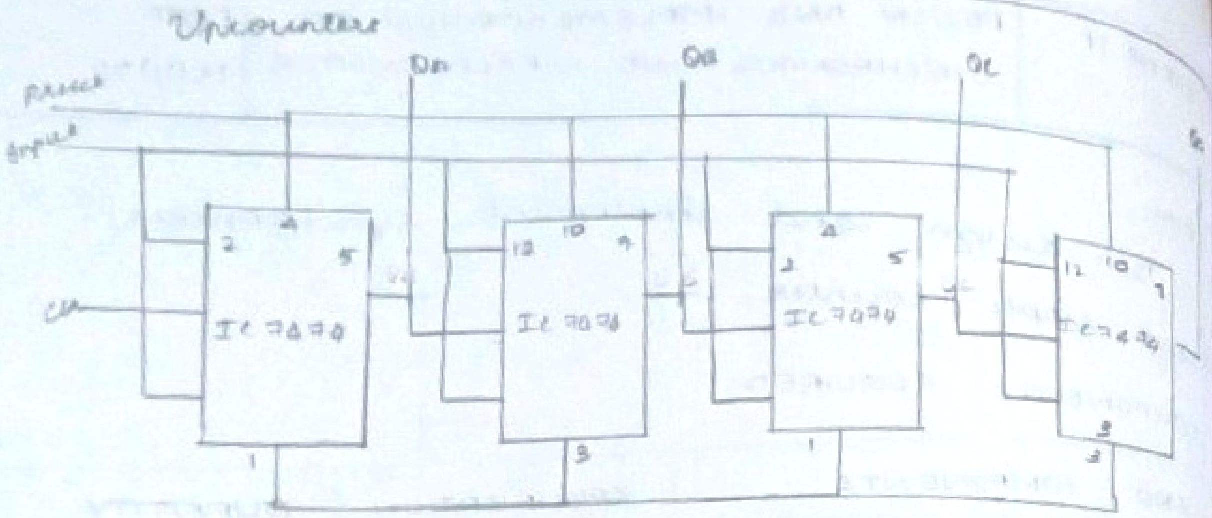
AIM:

To design and implement synchronous and ripple counter.

COMPONENTS REQUIRED:

S.NO	COMPONENTS	SPECIFICATION	QUANTITY
1	JK flip flop	IC 7473	1
2	AND gate	IC 7408	1
3	NAND gate	IC 7400	1
4	OR gate	IC 7432	1
5	X-OR gate	IC 7486	1
6	NOT gate	IC 7404	1
7	Digital trainer kit	-	1
8	Connecting wires.	-	As required

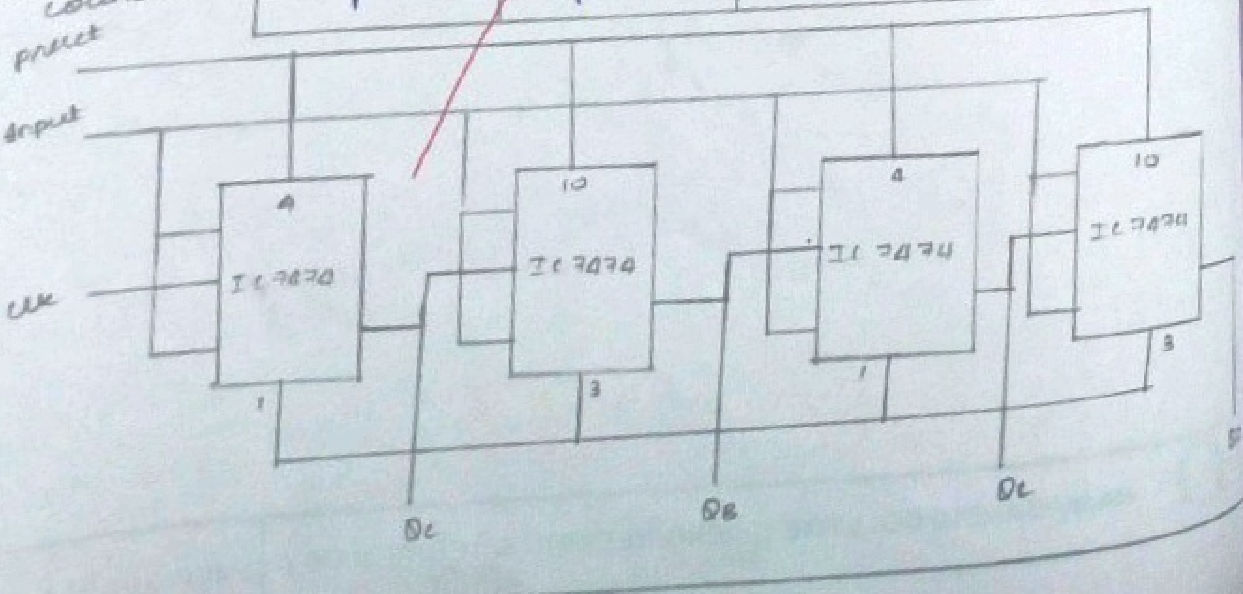
Upcounter



Output:

QA	QB	QC	QD
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Down counter



PROCEDURE:

* Connections are given as per the circuit diagram

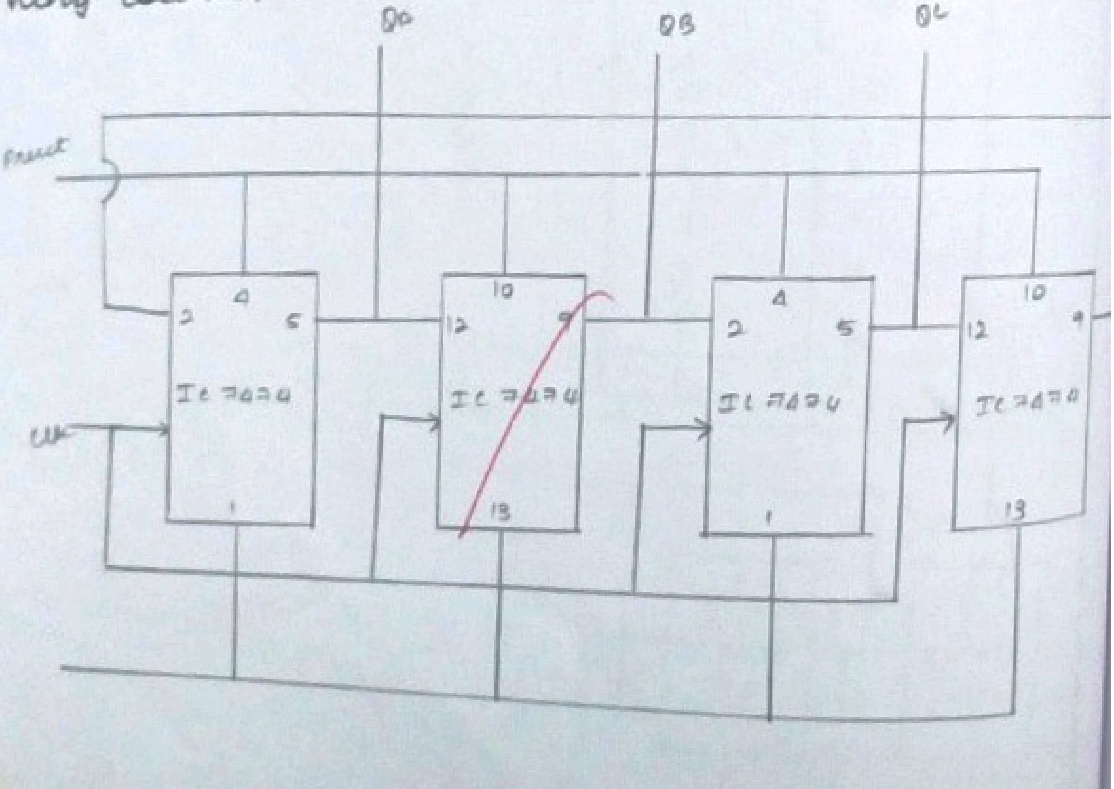
* Logical inputs are given as per the circuit diagram

* Observe the output and verify the truth table.

Output:

QA	QB	QC	QD
1	1	1	1
1	1	1	0
1	~~0~~1	0	1
1	1	0	0
1	0	1	~~0~~1
1	0	1	0
1	0	0	1
1	0	0	0
0	1	1	1
0	1	1	0
0	1	0	1
0	1	0	0
0	0	1	1
0	0	1	0
0	0	0	1

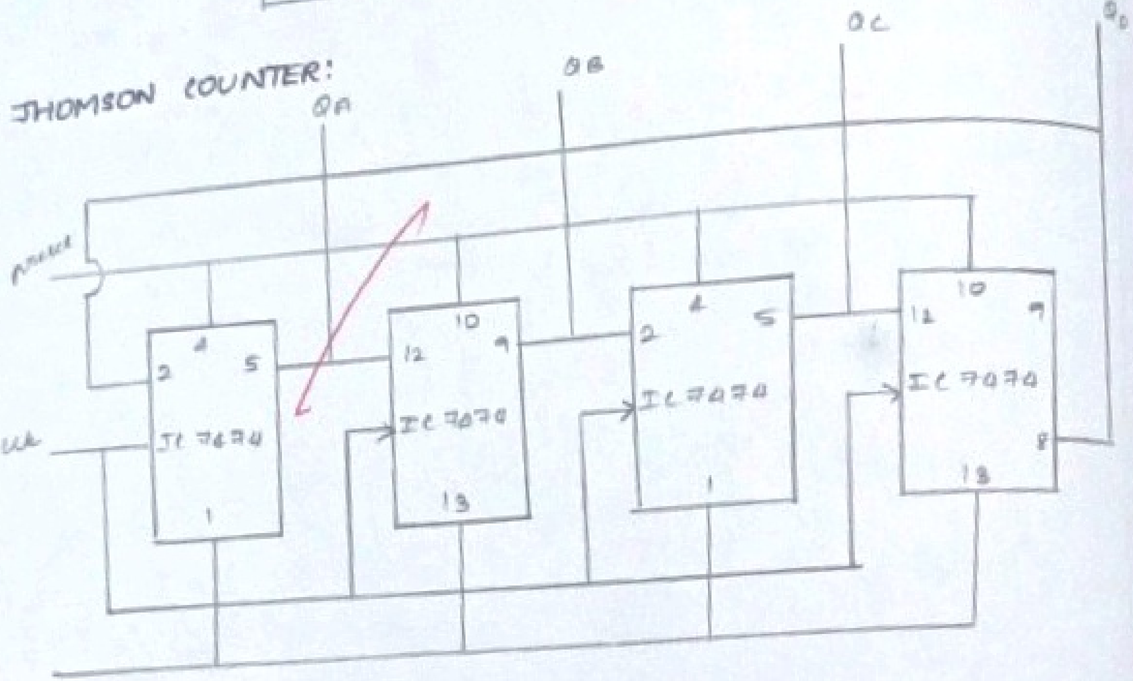
Ring Counter:



Output :

Clk	QA	QB	QC	QD
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

JHOMSON COUNTER:



Output :

Clk	QA	QB	QC	QD
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

RESULT:

Thus the design and implementation of synchronous and ripple counter was successfully verified.

EXP NO: 12	DESIGN AND IMPLEMENTATION OF SERIAL IN PARALLEL OUT & SERIAL IN SERIAL OUT	DATE: 22/11/22

AIM:

To design and implement,

(i) Serial In parallel out

(ii) Serial In serial out

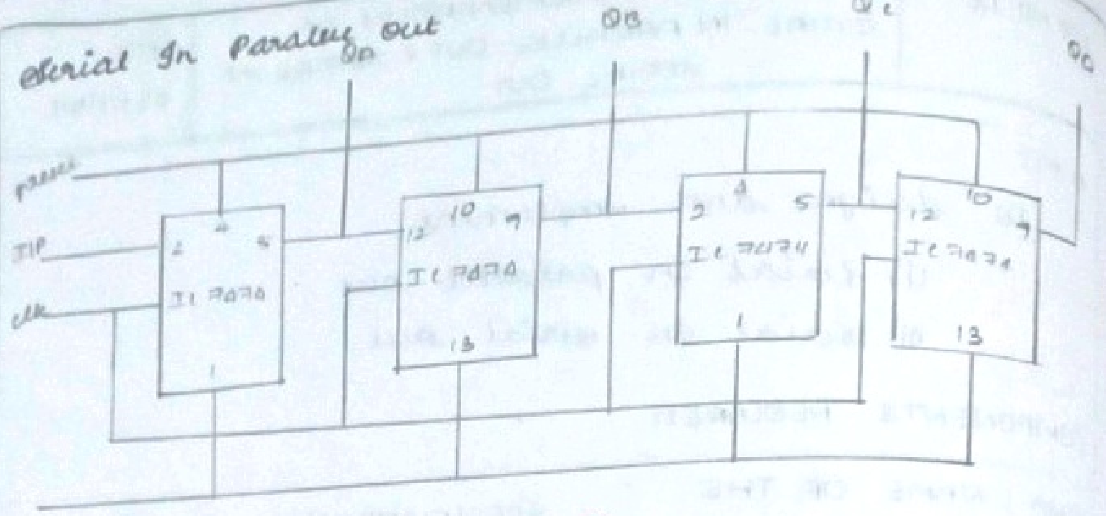
COMPONENTS REQUIRED:

S.NO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	D flip flop	IC 7474	2
2	OR gate	IC 7432	1
3	IC trainer kit	-	1
4	Path chord	-	As required.

THEORY:

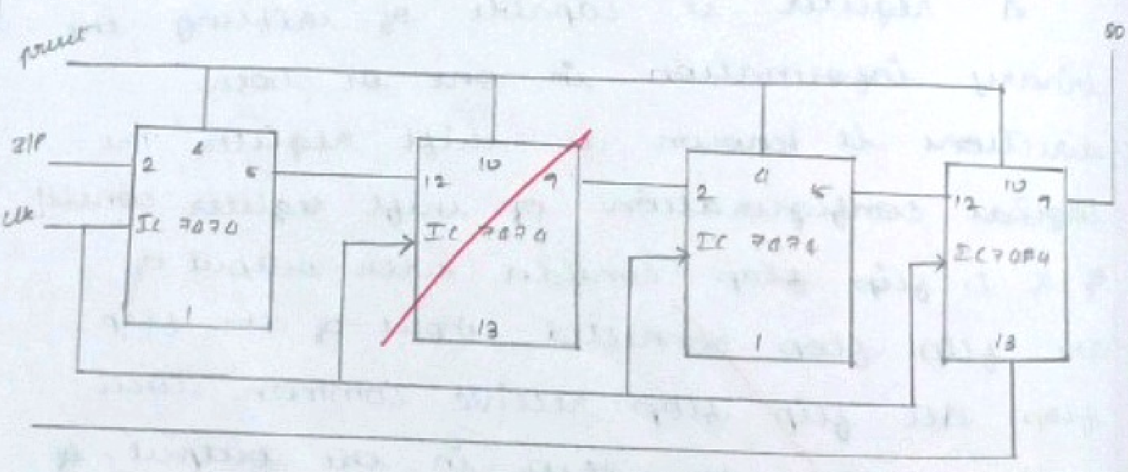
A register is capable of shifting the binary information in one or both directions is known as shift register. The logical configuration of shift register consist of a D-flip flop. Consider with output of the flip flop connected input of the flip flop. All flip flop receive common clock pulse which the shift in the output of the flip flop. The output of the flip flop is connected to the input of flip flop of the register. Each clock pulse shifts the

Serial In Parallel out



Clk	QA	QB	Qc	QD
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

Serial In Serial out



Clk	QA	QB	QC	QD
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	1

content of register one bit pointer to register.

PROCEDURE:

(*) Connections are given as per the circuit diagram

(*) Logical inputs are given as per the circuit diagram

(*) Observe the output and verify the truth table.

RESULT:

Thus the Implementation of shift register was successfully obtained.

EXP NO:13	IMPLEMENTATION OF SEQUENTIAL CIRCUIT USING PAL REALIZATION	DATE: 22/11/22

AIM:

To design and implement the sequential circuit using PLA realization.

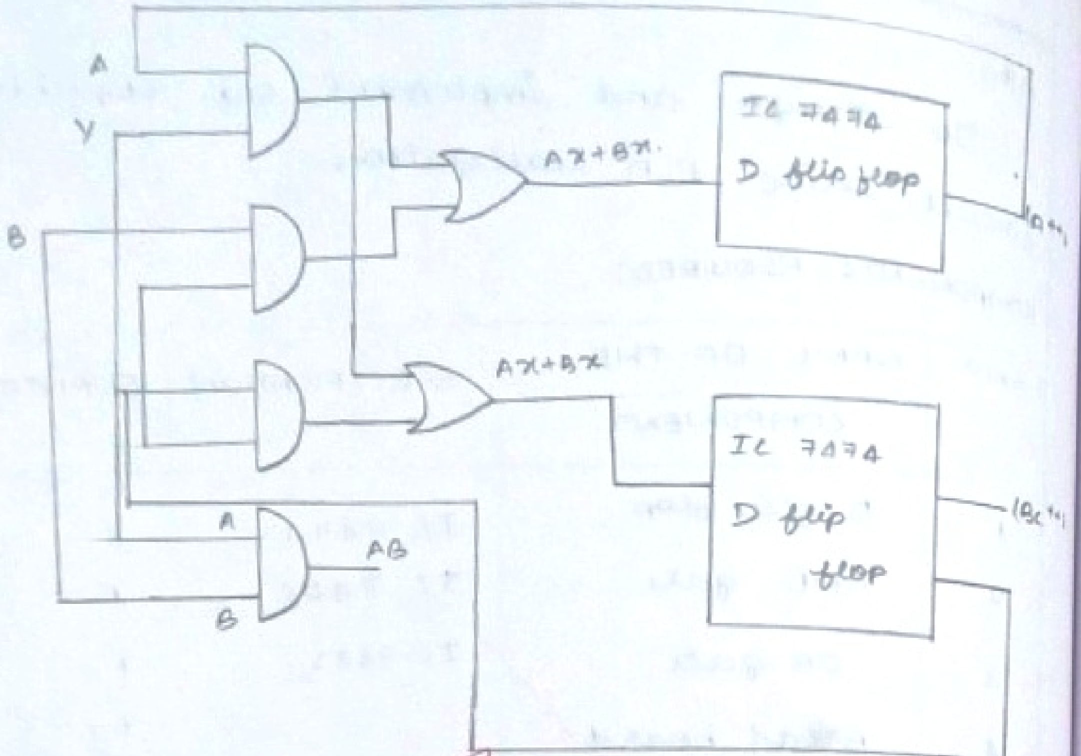
COMPONENTS REQUIRED:

SNO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	D flip flop	IC 7474	1
2	AND gate	IC 7408	1
3	OR gate	IC 7432	1
4	Bread board	-	1
5	IC trainer kit	-	1
6	Path chord	-	As required

THEORY:

The implementation of sequential circuit using PAL realization. D flip flop is a sequence of parallel input to the flip flop. The D flip flop gets the D-input from the OR gate output and to the output of D-flip flop is again as the input to AND gate which acts as a input to OR gate of D-flip flop.

LOGIC DIAGRAM:



TRUTH TABLE:

Minterm	Input					Output
	A	B	X	(A++)	(B++)	Y
Ax	1	-	1	1	1	-
Bx	-	1	1	1	-	-
B'x	-	0	1	-	1	-
AB	1	1	-	-	-	1

Present state			Input	Next state
A	B	X	A[t+1]	B(t+1)
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	0	0
1	1	1	1	1

PROCEDURE

(*) Connections are given as per the circuit diagram.

(*) Logical inputs are given as per the circuit diagram

(*) Observe the output and verify the truth table.

RESULT:

Thus the design and implement the sequential circuit using PLA realized is verified successfully.

EXP NO: 14	IMPLEMENTATION OF GIVEN BOOLEAN EXPRESSIONS USING MULTIOUTPUT PLA/PAL REALIZATION	DATE: 29/11/22

AIM:

To design and implement PLA realization for given boolean expression using multi output.

COMPONENTS REQUIRED:

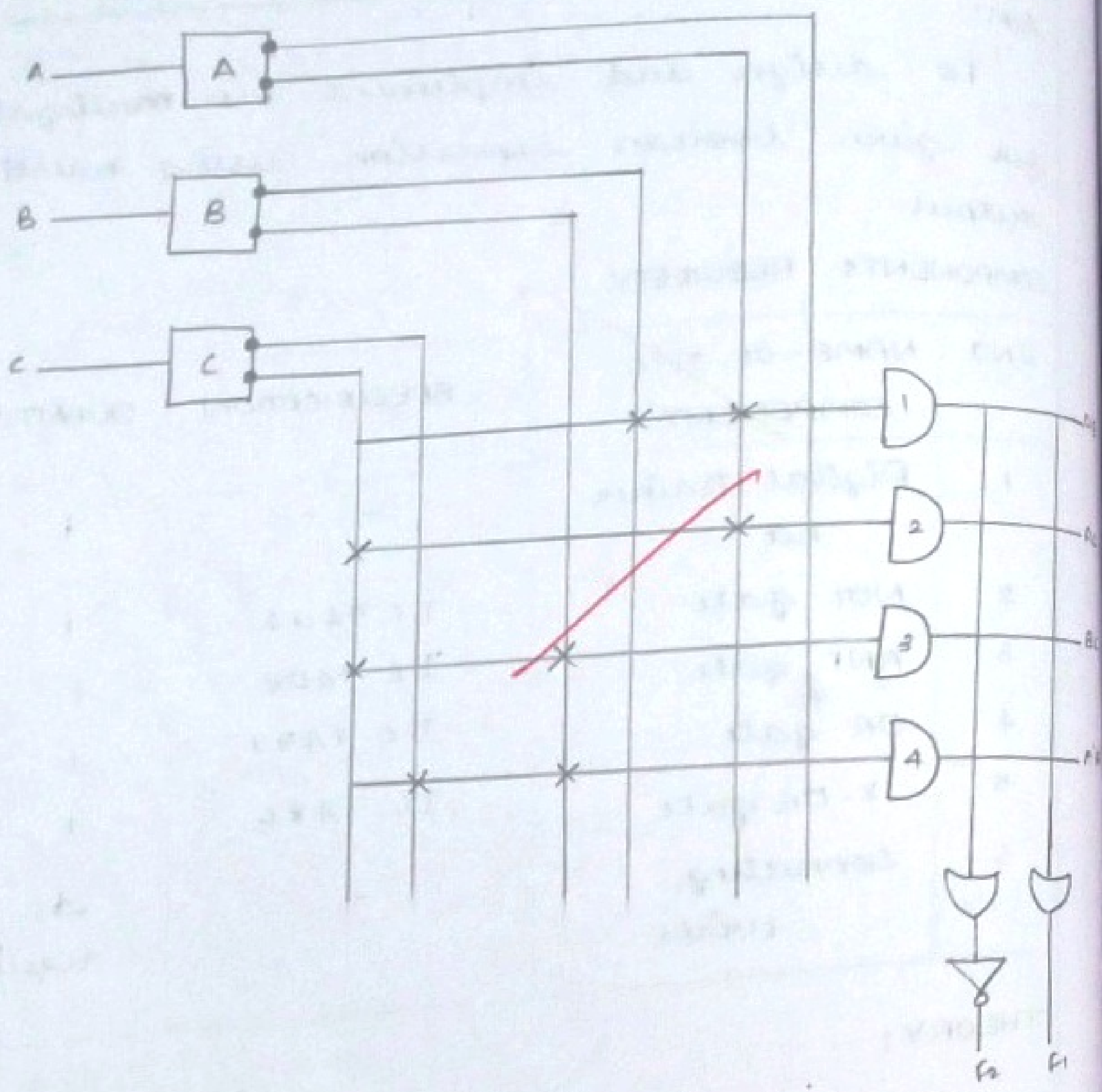
SNO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	Digital trainer kit	–	1
2	NOT gate	IC 7404	1
3	AND gate	IC 7408	1
4	OR gate	IC 7432	1
5	X-OR gate	IC 7486	1
6	Connecting wires	–	As required

THEORY:

PLA - Programmable logic array.

Programmable logic array (PLA) consists of two gate namely OR gate and AND gate. In programmable array logic they lies difference between PLA and PAL. This difference is that there in PAL only the AND array is programmable but OR array is fixed along AND array be array or be programmable based on the output.

LOGIC DIAGRAM:



TRUTH TABLE:

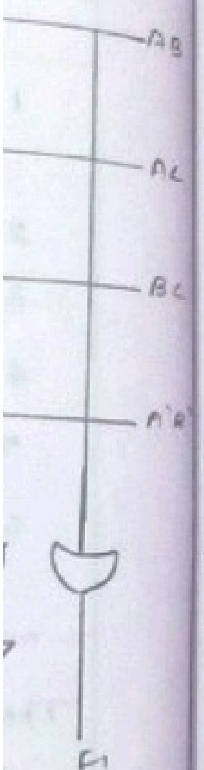
Product term		Input A	Input B	Input C	Output F1
AB'	1	1	0	–	1
AC	2	1	–	1	1
BC	3	–	1	1	–
A'BC'	4	0	1	0	1

PROCEDURE:

(*) Connections are made as per the given circuit diagram

(*) Logical input are given as per the circuit diagram.

(*) Observe the output and verify the truth table.



RESULT:

Thus the design and implementation of PAL realization for given boolean expression using multioutput is verified.

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